

CHAPTER 11 – PRACTICE SESSION

Please prepare your notebook; our practice session is about to start.



At the end of the lesson, the teacher will check your work.



5 mins

EXERCISE 1 (REF. P.254, EX.13)



The coefficient of x^3 in the expansion of $(3 + ax)^5$ is 12 times the coefficient of x^2 in the expansion of $\left(1 + \frac{ax}{2}\right)^4$. Find the value of a .



5 mins

EXERCISE 1 (REF. P.254, EX.13) – **ANS**



The coefficient of x^3 in the expansion of $(3 + ax)^5$ is 12 times the coefficient of x^2 in the expansion of $\left(1 + \frac{ax}{2}\right)^4$. Find the value of a .

$$\binom{5}{3} 3^2 (ax)^3 = 90a^3x^3, \quad \binom{4}{2} 1^2 \left(\frac{ax}{2}\right)^2 = \frac{3}{2}a^2x^2$$

$$a = \frac{1}{5} \text{ (or } a = 0\text{)}$$



6 mins

EXERCISE 2 (REF. P.278, EX.3)



a. i. Use the Binomial theorem to expand $(a + b)^4$, giving each term in its simplest form. [2]

ii. Hence find the term independent of x in the expansion of $\left(2x + \frac{1}{5x}\right)^4$. [2]

b. The coefficient of x^3 in the expansion of $\left(1 + \frac{x}{2}\right)^n$ equals $\frac{5n}{12}$. Find the value of the positive integer n . [3]



6 mins

EXERCISE 2 (REF. P.278, EX.3) – **ANS**



a. i. Use the Binomial theorem to expand $(a + b)^4$, giving each term in its simplest form. [2]

ii. Hence find the term independent of x in the

ex **a. i.** $a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$ [2]

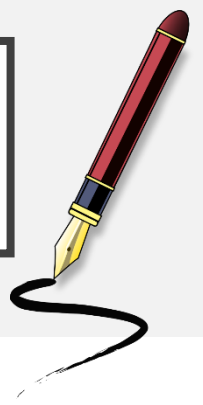
b. **ii.** $\frac{24}{25}$ [3]

$\frac{5n}{12}$
b. $n = 6, (n \geq 0)$



8 mins

EXERCISE 3 (REF. P.278, EX.8)

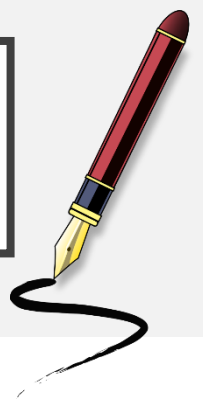


The first 3 terms in the expansion of $(a + x)^3 \left(1 - \frac{x}{3}\right)^5$, in ascending powers of x , can be written in the form $27 + bx + cx^2$, where a , b and c are integers. Find the values of a , b and c . [8]



8 mins

EXERCISE 3 (REF. P.278, EX.8) – **ANS**



The first 3 terms in the expansion of $(a + x)^3 \left(1 - \frac{x}{3}\right)^5$, in ascending powers of x , can be written in the form $27 + bx + cx^2$, where a , b and c are integers. Find the values of a , b and c . [8]

$$a = 3, \quad b = -18, \quad c = -6$$



6 mins

EXERCISE 4 (REF. PB P.120-121, EX.6, 8)



6. The sum of the first ten terms in an arithmetic progression is 120 and the sum of the first twenty terms is 840. Find the sum of the first thirty terms of this progression.

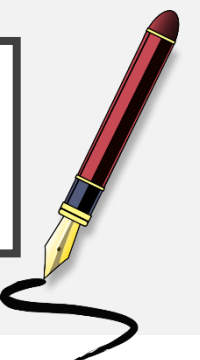
8. The sum of the first ten terms of an arithmetic progression is 240. The fifth term is 23. Find

- a.** the first term and the common difference of the progression
- b.** the sum of the first 60 terms of the series.



6 mins

EXERCISE 4 (REF. PB P.120-121, EX.6, 8)



6. The sum of the first ten terms in an arithmetic progression is 120 and the sum of the first twenty terms is 840. Find the sum of the first thirty terms of this progression.

$$\mathbf{6. \quad S_{30} = 1260}$$

8. The sum of the first ten terms of an arithmetic progression is 240. The fifth term is 23. Find

a. the first term and the common difference

b. the sum of the first 60 terms of the progression

$$\mathbf{8. \quad a. \quad a = 15, d = 2}$$

$$\mathbf{b. \quad S_{60} = 4440}$$



6 mins

EXERCISE 5 (REF. PB P.121, EX.15, 16)



15. The sum of the first n terms, S_n , of a particular arithmetic progression is given by $S_n = 2n^2 - n$. Find the sum of the terms from the third to the 12th inclusive.

16. The first, second and third terms of an arithmetic progression are $8 - x$, $3x$ and $4x + 1$, respectively. Calculate the value of x and find the sum of the first eight terms of the progression.



6 mins

EXERCISE 5 (REF. PB P.121, EX.15, 16)



15. The sum of the first n terms, S_n , of a particular arithmetic progression is given by $S_n = 2n^2 - n$. Find the sum of the terms from the third to the 12th inclusive.

15. 270

16. The first, second and third terms of an arithmetic progression are $8 - x$, $3x$ and $4x + 1$, respectively. Calculate the value of x and find the sum of the first eight terms of the progression.

16. $x = 3, S_8 = 152$



6 mins

EXERCISE 6 (REF. PB P.125, EX.11.4.11 AND 11.4.14)



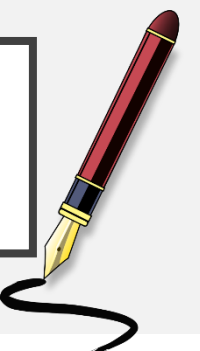
11. Which is the first term of the geometric progression $5, 10, 20, \dots$ to exceed 400 000?

14. How many terms of the geometric progression $5, -10, +20, - \dots$ should be chosen so that the sum equals 215?



6 mins

EXERCISE 6 (REF. PB P.125, EX.11.4.11 AND 11.4.14)



11. Which is the first term of the geometric progression 5, 10, 20, ... to exceed 400 000?

18th term, or $n = 18$

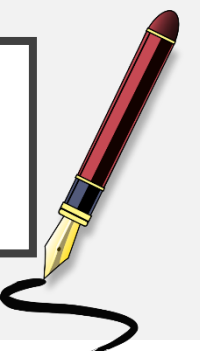
14. How many terms of the geometric series 5, -10 , $+20$, $-\dots$ should be chosen so that the sum equals 215?

7 terms



6 mins

EXERCISE 7 (REF. PB P.125, EX.11.4.17 AND 11.4.18)



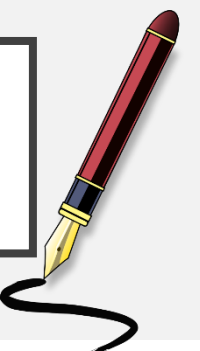
17. Find the values of x given that 4, x , and $x^2 - 1$ are consecutive geometric terms.

18. Find the first term of the geometric sequence $24, 8, \frac{8}{3}, \frac{8}{9}, \dots$ which is less than 0.001



6 mins

EXERCISE 7 (REF. PB P.125, EX.11.4.17 AND 11.4.18)



17. Find the values of x given that 4, x , and $x^2 - 1$ are consecutive geometric terms.

$$\pm \frac{2\sqrt{3}}{3}$$

18. Find the first term of the geometric sequence $24, 8, \frac{8}{3}, \frac{8}{9}, \dots$ which is less than 0.001

$$\text{The } 11^{\text{th}} \text{ term, } \left(\frac{8}{19\,683} \right)$$



4 mins

EXERCISE 8 (REF. PB P.125, EX.11.4.20)



20. Find the first five terms of a geometric series such that the sum of its first and third terms is 50 and the sum of the second and fourth terms is 150.



4 mins

EXERCISE 8 (REF. PB P.125, EX.11.4.20)



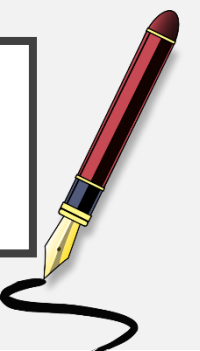
20. Find the first five terms of a geometric series such that the sum of its first and third terms is 50 and the sum of the second and fourth terms is 150.

$$\begin{aligned} a_1 &= 5, & a_2 &= 15, & a_3 &= 45, \\ a_4 &= 135, & a_5 &= 405 \end{aligned}$$



4 mins

EXERCISE 9 (REF. PB P., EX.11.5.11)



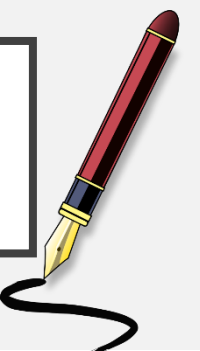
11. The first three terms of a geometric progression are $k - 8$, $k + 4$ and $3k + 2$ respectively, where k is a positive constant. Find

- a)** the value of k
- b)** the common ratio
- c)** explain why it is not possible to find the sum to infinity for this progression



4 mins

EXERCISE 9 (REF. PB P., EX.11.5.11)



11. The first three terms of a geometric progression are $k - 8$, $k + 4$ and $3k + 2$ respectively, where k is a positive constant. Find

- a)** the value of k
- b)** the common ratio
- c)** explain why it is not a geometric progression for this progression

a) $k = 16$

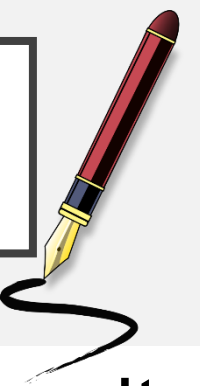
b) $r = \frac{5}{2}$

c) Because r is not in the range $-1 < r < 1$ or $|r| < 1$



5 mins

EXERCISE 10 (REF. P.273 EX.15, PB P. EX.11.5.21)



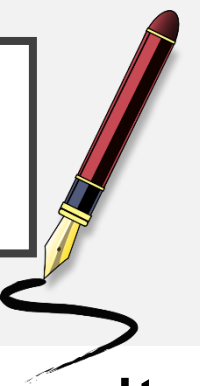
15. A ball is dropped from a height of 12 m. After each bounce it rises to $\frac{3}{4}$ of the height of the previous bounce. Find the total vertical distance that the ball travels.

21. An insect walks 0.6 m east and then 0.36 m west, 0.216 m east and so on. Find its final position and the total distance it has walked to get there.



5 mins

EXERCISE 10 (REF. P.273 EX.15, PB P. EX.11.5.21)



15. A ball is dropped from a height of 12 m. After each bounce it rises to $\frac{3}{4}$ of the height of the previous bounce. Find the total vertical distance that the ball travels.

15. 84 m

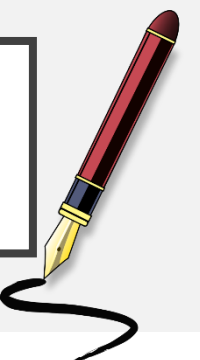
21. An ins
east and so
walked to g

21. 0.375 m east of its starting point, 1.5 m



5 mins

EXERCISE 11 (REF. PB P. EX.11.6.2-11.6.3)



2. The second, fifth and 11th terms of an arithmetic progression are in geometric progression and the seventh term is 4.

a. the first term and the common difference

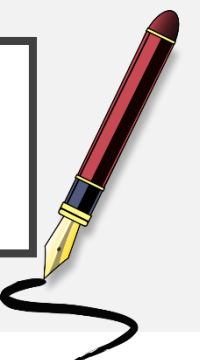
b. The common ratio of the geometric progression

3. The second, third and sixth terms of an arithmetic progression are consecutive terms of a geometric progression. Find the common ratio of the geometric progression.



5 mins

EXERCISE 11 (REF. PB P. EX.11.6.2-11.6.3)



- 2.** The second, fifth and 11th terms of an arithmetic progression are in geometric progression and the seventh term is 4.
- a.** the first term and the common difference
 - b.** The common ratio of the geometric progression

2. a. $a = 1, d = \frac{1}{2}$

b. $r = 2$

3. $r = 3$

- 3.** The second, third and sixth terms of an arithmetic progression are consecutive terms of a geometric progression with a common ratio of the geometric progression.